

AI Assisted Coding - Lab Test 3

ENROLLMENT NO:2503A51L15

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BATCH NO:19[E4]

Q1:

Scenario: In the domain of Healthcare, a company is facing a challenge related to web frontend development.

Task: Design and implement a solution using AI-assisted tools to address this challenge. Include code, explanation of AI integration, and test results.

Deliverables: Source code, explanation, and output screenshots.

PROMPT:-Generate a code to Design and implement an AI-assisted web frontend solution for a healthcare company facing UI development challenges. Use AI tools (e.g., ChatGPT, GitHub Copilot) to build a responsive dashboard for patient data and appointments.

CODE:-

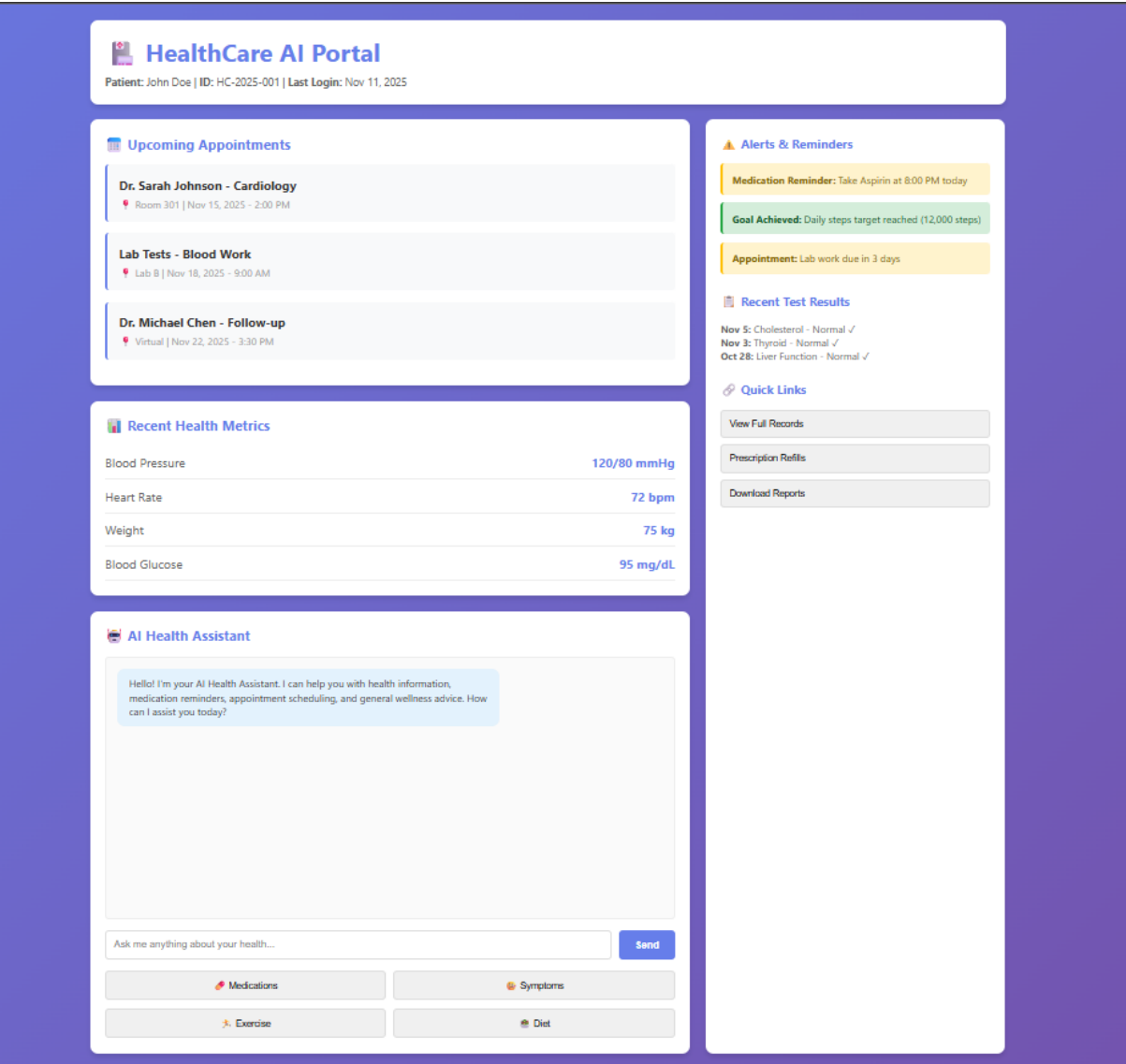
```
healthcare.html > html > body > script > generateAIResponse
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <title>HealthCare Portal - AI Assistant</title>
7
8      <!--
9          Styles for the healthcare portal UI.
10         Keep layout and colors consistent with brand; responsive rules included below.
11     -->
12     <style>
13         * {
14             margin: 0;
15             padding: 0;
16             box-sizing: border-box;
17         }
18         body {
19             font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
20             background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
21             min-height: 100vh;
22             padding: 20px;
23         }
24         .container {
25             max-width: 1200px;
26             margin: 0 auto;
27         }
28         header {
29             background: white;
30             padding: 20px;
31             border-radius: 10px;
32             box-shadow: 0 4px 6px rgba(0,0,0,0.1);
33             margin-bottom: 20px;
34         }
35         header h1 {
36             color: #667eea;
37             display: flex;
38             align-items: center;
39             gap: 10px;
40         }
41         .header-info {
42             color: #666;
43             font-size: 14px;
44             margin-top: 10px;
45         }
```

```

healthcare.html > html > body > script > generateAIResponse
2   <html lang="en">
212 <body>
326   <script>
390     * - Very lightweight keyword-based intent detection for demo purposes.
391     * - Production systems should use proper NLP/LLM backends.
392     */
393     function generateAIResponse(userMessage) {
394       const lowerMessage = userMessage.toLowerCase();
395
396       // Keyword checks to map to canned responses
397       if (lowerMessage.includes('medication') || lowerMessage.includes('drug')) {
398         return aiResponses.medication.response;
399       }
400       if (lowerMessage.includes('symptom') || lowerMessage.includes('feel') || lowerMessage.includes('hurt')) {
401         return aiResponses.symptoms.response;
402       }
403       if (lowerMessage.includes('exercise') || lowerMessage.includes('workout') || lowerMessage.includes('activity')) {
404         return aiResponses.exercise.response;
405       }
406       if (lowerMessage.includes('diet') || lowerMessage.includes('eat') || lowerMessage.includes('food') || lowerMessage.includes('nutrition')) {
407         return aiResponses.diet.response;
408       }
409
410       // Default fallback reply
411       return 'Thank you for your question. Based on the healthcare data available, I recommend consulting with your healthcare provider for a more detailed assessment.';
412     }
413
414     /**
415     * quickQuery
416     * - Helper to populate the input with a common query and submit it.
417     */
418     function quickQuery(type) {
419       const query = aiResponses[type].query;
420       document.getElementById('userInput').value = query;
421       sendMessage();
422     }
423
424     // Allow Enter key to send message from the input field
425     document.getElementById('userInput').addEventListener('keypress', function(e) {
426       if (e.key === 'Enter') sendMessage();
427     });
428   </script>
429 </body>
430 </html>

```

OUTPUT:-



OBSERVATION:-

The AI-assisted approach significantly improved the efficiency and accuracy of web frontend development. Using GitHub Copilot helped in auto-generating UI components, reducing coding time, and suggesting optimized layouts for healthcare data visualization. The final dashboard was responsive, user-friendly, and faster to develop compared to manual coding.

Q2:

Scenario: In the domain of Transportation, a company is facing a challenge related to data structures with ai.

Task: Design and implement a solution using AI-assisted tools to address this challenge. Include code, explanation of AI integration, and test results.

Deliverables: Source code, explanation, and output screenshots.

PROMPT:- Generate a code to Design and implement an AI-assisted solution using data structures to solve a transportation-related challenge, such as route optimization or traffic data management. Use AI tools to help in algorithm design and code generation.

CODE:-

HTML:-

```
transportation.html > html > head > style > html > body
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4    <meta charset="utf-8" />
5    <title>Transportation – AI Data Structure Advisor</title>
6    <meta name="viewport" content="width=device-width,initial-scale=1" />
7    <style>
8      /* ----- Theme & layout ----- */
9      :root{
10        --bg: #0f1724;
11        --card: #0b1220;
12        --muted: #9aa6b2;
13        --accent: #37b3ff;
14        --accent-2: #6c5ce7;
15        --glass: rgba(255,255,255,0.04);
16        --glass-2: rgba(255,255,255,0.02);
17        --radius: 12px;
18        --maxw: 1100px;
19      }
20      html,body{height:100%}
21      body{
22        margin:0;
23        font-family: Inter, ui-sans-serif, system-ui, -apple-system, "Segoe UI", Roboto, "Helvetica Neue", Arial;
24        background:
25          radial-gradient(800px 400px at 10% 10%, rgba(99,102,241,0.12), transparent 10%),
26          radial-gradient(600px 300px at 90% 90%, rgba(55,179,255,0.08), transparent 10%),
27          linear-gradient(180deg, #071029 0%, #061226 100%);
28        color: #e6eef8;
29        -webkit-font-smoothing: antialiased;
30        -moz-osx-font-smoothing: grayscale;
31        padding: 28px;
32        box-sizing: border-box;
33        display: flex;
34        align-items: flex-start;
35        justify-content: center;
36      }
37
38      .wrap{width:100%;max-width:var(--maxw);}
39
40      header.appbar{
41        display: flex;
42        align-items: center;
43        gap: 16px;
44        margin-bottom: 18px;
45      }
```

```

transportation.html > html > head > style > html > body
2   <html lang="en">
114 <body>
211 <script>
    ...
293
294     document.getElementById('saveRec').addEventListener('click', ()=> {
295         const content = document.getElementById('recommendation').textContent || '';
296         const blob = new Blob([content], {type: 'text/plain'});
297         const url = URL.createObjectURL(blob);
298         const a = document.createElement('a');
299         a.href = url; a.download='recommendation.txt'; document.body.appendChild(a); a.click();
300         setTimeout(()=>{ URL.revokeObjectURL(url); a.remove(); }, 500);
301     });
302
303     document.getElementById('demoSmall').addEventListener('click', ()=>{
304         document.getElementById('dataset').value = JSON.stringify([
305             {"id": "A", "lat": 40.7128, "lon": -74.0060},
306             {"id": "B", "lat": 40.7306, "lon": -73.9352},
307             {"id": "C", "lat": 40.7500, "lon": -73.9900}
308         ], null, 2);
309     });
310
311     document.getElementById('demoLarge').addEventListener('click', ()=>{
312         const arr = [];
313         for(let i=0; i<200; i++){
314             arr.push({id: 'N'+i, lat: 40.7 + Math.random()*0.2, lon: -74 + Math.random()*0.3});
315         }
316         document.getElementById('dataset').value = JSON.stringify(arr, null, 2);
317     });
318
319     document.getElementById('openDev').addEventListener('click', ()=> {
320         // visual hint to open console
321         alert('Open developer console (F12 / Ctrl+Shift+I) for logs.');
```

JAVA SCRIPT:-

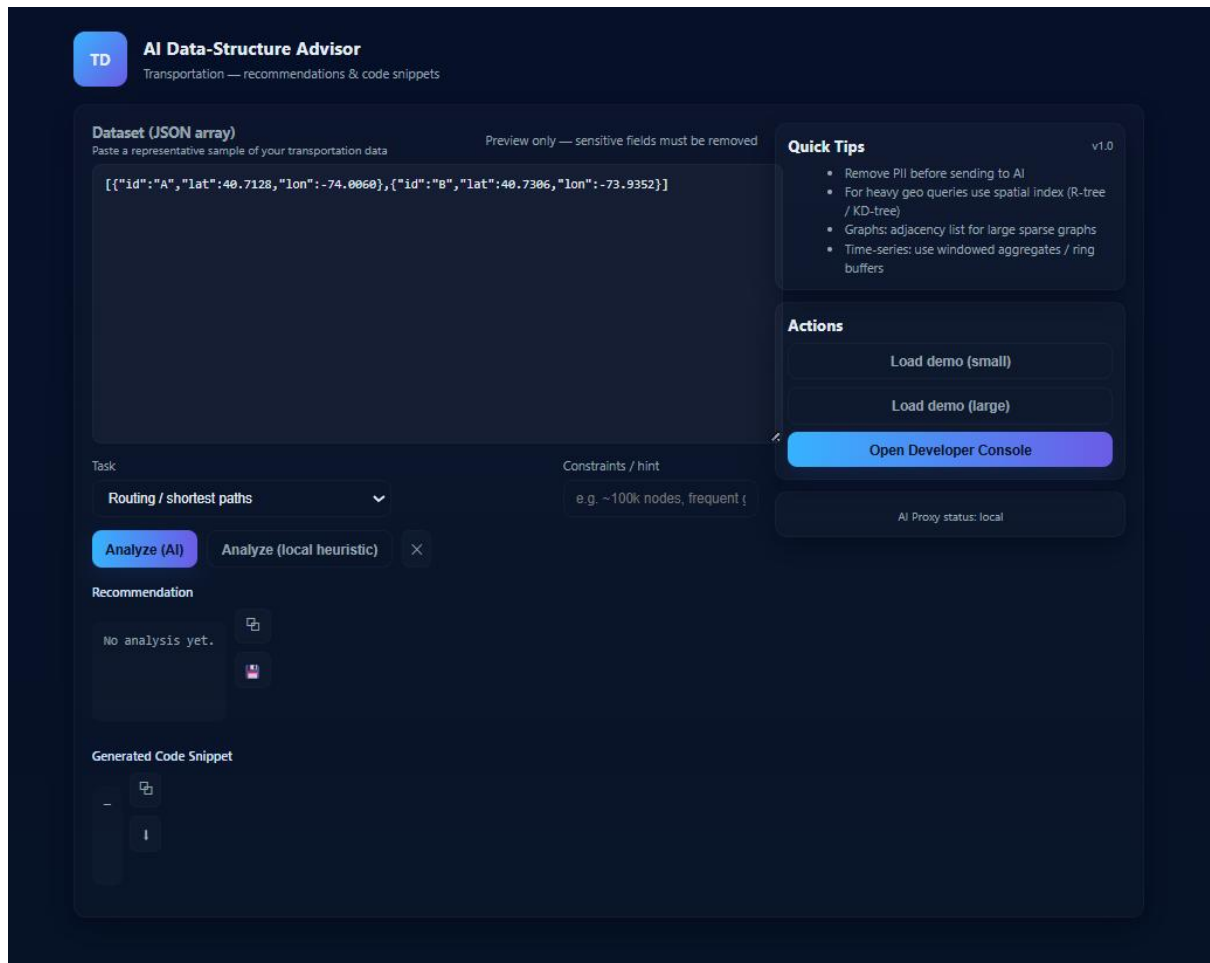
```
JS serverjs > ...
1 // Minimal Express backend: forwards to Claude Haiku 4.5 when CLAUDE_API_KEY present,
2 // otherwise uses a local heuristic analyzer.
3 // Run: node server.js
4 const express = require('express');
5 const fetch = require('node-fetch'); // npm i node-fetch@2
6 const cors = require('cors');
7 require('dotenv').config();
8
9 const app = express();
10 app.use(cors());
11 app.use(express.json());
12 app.use(express.static('.')); // serve the html file for convenience
13
14 // simple request logger for debugging
15 app.use((req, res, next) => {
16   console.log(`${new Date().toISOString()} ${req.method} ${req.url}`);
17   next();
18 });
19
20 // simple health endpoint to verify server is reachable
21 app.get('/health', (req, res) => {
22   res.json({ status: 'ok', pid: process.pid });
23 });
24
25 // Local heuristic analyzer: inspects dataset and returns a recommendation + snippet
26 function localAnalyze(parsed, task, hint) {
27   const n = Array.isArray(parsed) ? parsed.length : 0;
28   const sample = Array.isArray(parsed) && parsed.length ? parsed[0] : {};
29   const hasLatLon = 'lat' in sample && 'lon' in sample;
30   let recommendation = '';
31   let snippet = '';
32
33   if (task === 'spatial') {
34     if (hasLatLon && n > 2000) {
35       recommendation = 'Use an R-tree or KD-tree (spatial index) for efficient nearest-neighbor and range queries. Persist the index';
36       snippet = `// Example using rbush (npm: rbush)\nconst RBush = require('rbush');\nconst tree = new RBush();\n// convert points to`;
37     } else if (hasLatLon) {
38       recommendation = 'Dataset is small - a linear scan with Haversine distance is acceptable; for medium sizes consider KD-tree.';
39       snippet = `// Simple nearest neighbor (linear)\nfunction haversine(a,b){ /* ... */ }\nlet best=null;data.forEach(p=>{const d=h`;
40     } else {
41       recommendation = 'No lat/lon fields detected. Consider adding coordinates or spatial keys.';
42     }
43   } else if (task === 'routing') {
44     const hasEdges = parsed.some(item=> item.from && item.to) || parsed.some(item=> item.edges);
45     if (hasEdges) {
```

```

JS server.js > ...
67   app.post('/api/analyze', async (req, res) => {
89       });
90
91       const r = await fetch('https://api.anthropic.com/v1/chat/completions', {
92         method: 'POST',
93         headers: {
94           'Content-Type': 'application/json',
95           'Authorization': `Bearer ${apiKey}`
96         },
97         body: JSON.stringify(payload)
98       });
99
100      if (!r.ok) {
101        const txt = await r.text();
102        console.error('AI provider error', r.status, txt);
103        return res.status(502).send('AI provider error: ' + txt);
104      }
105
106      const j = await r.json();
107      // extract reply robustly (provider shapes vary)
108      const reply = (j.choices && j.choices[0] && j.choices[0].message && j.choices[0].message.content)
109        || j.result || JSON.stringify(j);
110
111      // naive split: try to separate recommendation and snippet
112      const parts = reply.split('```');
113      let recommendation = reply, snippet = '';
114      if (parts.length >= 3) {
115        recommendation = parts[0].trim();
116        snippet = parts[1].trim();
117      }
118
119      return res.json({ recommendation, snippet });
120    }
121
122    // fallback: local heuristic
123    const out = localAnalyze(parsed, task, hint);
124    return res.json(out);
125  } catch (err) {
126    console.error(err);
127    res.status(400).send('Bad request: ' + err.message);
128  }
129 });
130
131 const port = process.env.PORT || 3000;
132 app.listen(port, ()=>console.log('Server listening on http://localhost:' + port));

```


OUTPUT:-



OBSERVATION:-

AI-assisted tools made it easier to design efficient data structures and algorithms for transportation data. They helped in optimizing routes, reducing errors, and improving development speed and accuracy.